

7.5 离散数据的曲线拟合

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7.5 离散数据的曲线拟合

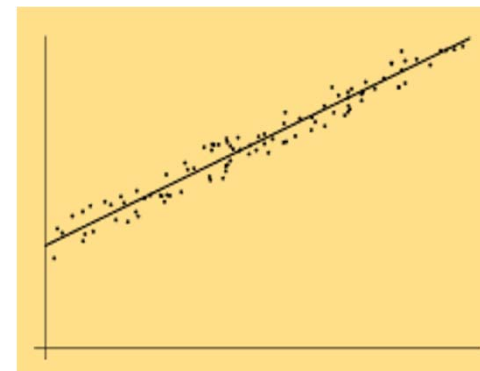
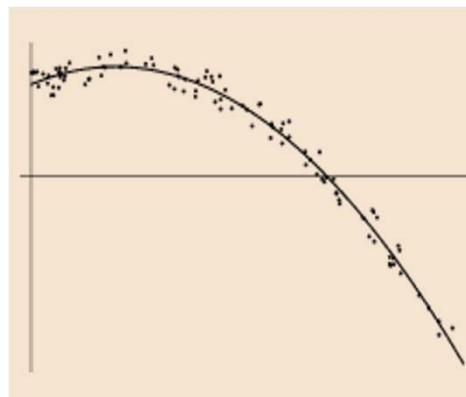
- 7.5.1 曲线拟合问题
- 7.5.2 多项式拟合
- 7.5.3 正交多项式拟合

7.5.1 曲线拟合问题

$$(x_i, y_i) (i = 0, 1, 2, \dots, n)$$

$$p(x)$$

$$\delta_i = p(x_i) - y_i$$
$$(i = 0, 1, 2, \dots, n)$$



$$(1) \quad \|\delta\|_1 = \sum_{i=0}^n |\delta_i|$$

$$(2) \quad \|\delta\|_2^2 = \sum_{i=0}^n \delta_i^2$$

$$(3) \quad \|\delta\|_\infty = \max_{0 \leq i \leq n} |\delta_i|$$

最小二乘法

称使范数 $\|\delta\|_2^2$ 达到最小的曲线拟合方法为曲线拟合的最小二乘法。

状态估计理论的起源，亦可追溯至早期的天文学。高斯（Gauss）最早¹提出了最小二乘法，并用于最小化观测误差对行星估计轨道的影响。据记载，在谷神星最后一次被观测到的九个月后，他用最小二乘法预测了谷神星出现在太阳背后的位置，误差仅在 0.5 度以内。那是 1801 年，高斯仅 23 岁。随后在 1809 年，他证明了在正态分布误差假设下，最小二乘解即最优估计。大部分沿用至今的经典估计方法，都可以追溯到高斯的最小二乘法。

卡尔·弗里德里希·高斯（Carl Friedrich Gauss, 1777—1855），德国数学家，在很多领域都有重大贡献，包括统计学和估计理论。

¹ 有一些争议认为勒让德（Adrien Marie Legendre）比高斯更早地提出了最小二乘法。

摘自《机器人学中的状态估计》. 西安交通大学出版社. 2018.11, p2

BACK

7.5.2 多项式拟合

函数拟合的求解方法

对于给定的一组数据 (x_i, y_i) ($i = 0, 1, 2, \dots, n$) 在函数类 $\Phi = \{\varphi_0(x), \varphi_1(x), \dots, \varphi_m(x)\}$ 中寻求一个函数 $p(x)$, 使误差平方和:

$$\|\delta\|_2^2 = \sum_{i=0}^n \delta_i^2 = \sum_{i=0}^n (p(x_i) - y_i)^2 \quad (7.5.1)$$

达到最小

$$p(x) = a_0 \varphi_0(x) + a_1 \varphi_1(x) + \dots + a_m \varphi_m(x) \quad (m < n) \quad (7.5.2)$$

$$\|\delta\|_2^2 = \sum_{i=0}^n \delta_i^2 = \sum_{i=0}^n (p(x_i) - y_i)^2$$

$$p(x) = a_0\varphi_0(x) + a_1\varphi_1(x) + \dots + a_m\varphi_m(x)$$

使误差平方和 $\|\delta\|_2^2$ 取最小值问题, 可转化为

求多元函数:

$$F(a_0, a_1, \dots, a_m) = \sum_{i=0}^n \left(\sum_{j=0}^m a_j \varphi_j(x_i) - y_i \right)^2$$

的极小点 $(a_0^*, a_1^*, \dots, a_m^*)$

令:

$$\frac{\partial F}{\partial a_k} = 0 \quad (k = 0, 1, 2, \dots, m)$$

由此得:

$$\sum_{i=0}^n \left(\sum_{j=0}^m a_j \varphi_j(x_i) - y_i \right) \varphi_k(x_i) = 0 \quad (k = 0, 1, 2, \dots, m) \quad (7.5.3)$$

$$\sum_{i=0}^n \left(\sum_{j=0}^m a_j \varphi_j(x_i) - y_i \right) \varphi_k(x_i) = 0 \quad (k = 0, 1, 2, \dots, m) \quad (7.5.3)$$

记离散意义下函数的内积符号：

$$(\varphi_j, \varphi_k) = \sum_{i=0}^n \varphi_j(x_i) \varphi_k(x_i)$$

$$(f, \varphi_k) = \sum_{i=0}^n y_i \varphi_k(x_i) = d_k$$

则(7.5.3)式可写为：

$$\sum_{j=0}^m (\varphi_k, \varphi_j) a_j = d_k$$

$$(k = 0, 1, 2, \dots, m)$$

法方程组

$$\sum_{j=0}^n (\varphi_k, \varphi_j) a_j = d_k \quad (k = 0, 1, 2, \dots, m)$$

$$\begin{bmatrix} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_1) & \dots & (\varphi_0, \varphi_m) \\ (\varphi_1, \varphi_0) & (\varphi_1, \varphi_1) & \dots & (\varphi_1, \varphi_m) \\ \dots & \dots & \dots & \dots \\ (\varphi_m, \varphi_0) & (\varphi_m, \varphi_1) & \dots & (\varphi_m, \varphi_m) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_m \end{bmatrix} = \begin{bmatrix} d_0 \\ d_1 \\ \dots \\ d_m \end{bmatrix} \quad (7.5.4)$$

$$(\varphi_j, \varphi_k) = \sum_{i=0}^n \varphi_j(x_i) \varphi_k(x_i)$$

$$(f, \varphi_k) = \sum_{i=0}^n y_i \varphi_k(x_i) = d_k$$

$$\begin{bmatrix} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_1) & \dots & (\varphi_0, \varphi_m) \\ (\varphi_1, \varphi_0) & (\varphi_1, \varphi_1) & \dots & (\varphi_1, \varphi_m) \\ \dots & \dots & \dots & \dots \\ (\varphi_m, \varphi_0) & (\varphi_m, \varphi_1) & \dots & (\varphi_m, \varphi_m) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_m \end{bmatrix} = \begin{bmatrix} d_0 \\ d_1 \\ \dots \\ d_m \end{bmatrix} \quad (7.5.4)$$

$$(\varphi_j, \varphi_k) = \sum_{i=0}^n \varphi_j(x_i) \varphi_k(x_i)$$

$$(f, \varphi_k) = \sum_{i=0}^n y_i \varphi_k(x_i) = d_k$$

多项式拟合

当函数类 $\Phi = \{\varphi_0, \varphi_1, \dots, \varphi_m\} = \{1, x, x^2, \dots, x^m\}$ 时：

$$p(x) = a_0 + a_1 x + \dots + a_m x^m$$

$$\begin{bmatrix} \sum 1 & \sum x_i & \dots & \sum x_i^m \\ \sum x_i & \sum x_i^2 & \dots & \sum x_i^{m-1} \\ \dots & \dots & \dots & \dots \\ \sum x_i^m & \sum x_i^{m+1} & \dots & \sum x_i^{2m} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_m \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \dots \\ \sum x_i^m y_i \end{bmatrix} \quad (7.5.5)$$

求拟合多项式的步骤:

(1) 确定拟合多项式的次数。

根据给定数据点的分布, 确定次数。

(2) 求解法方程(7.5.5)式, 求出系数。

$$a_i^*, \quad (i = 0, 1, 2, \dots, n)$$

$$\begin{bmatrix} \sum 1 & \sum x_i & \dots & \sum x_i^m \\ \sum x_i & \sum x_i^2 & \dots & \sum x_i^{m-1} \\ \dots & \dots & \dots & \dots \\ \sum x_i^m & \sum x_i^{m+1} & \dots & \sum x_i^{2m} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_m \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \dots \\ \sum x_i^m y_i \end{bmatrix}$$

(3) 写出拟合多项式。

$$S(x) = \sum_{i=0}^n a_i x^i$$

$$\begin{bmatrix} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_1) & \dots & (\varphi_0, \varphi_m) \\ (\varphi_1, \varphi_0) & (\varphi_1, \varphi_1) & \dots & (\varphi_1, \varphi_m) \\ \dots & \dots & \dots & \dots \\ (\varphi_m, \varphi_0) & (\varphi_m, \varphi_1) & \dots & (\varphi_m, \varphi_m) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_m \end{bmatrix} = \begin{bmatrix} d_0 \\ d_1 \\ \dots \\ d_m \end{bmatrix}$$

举例

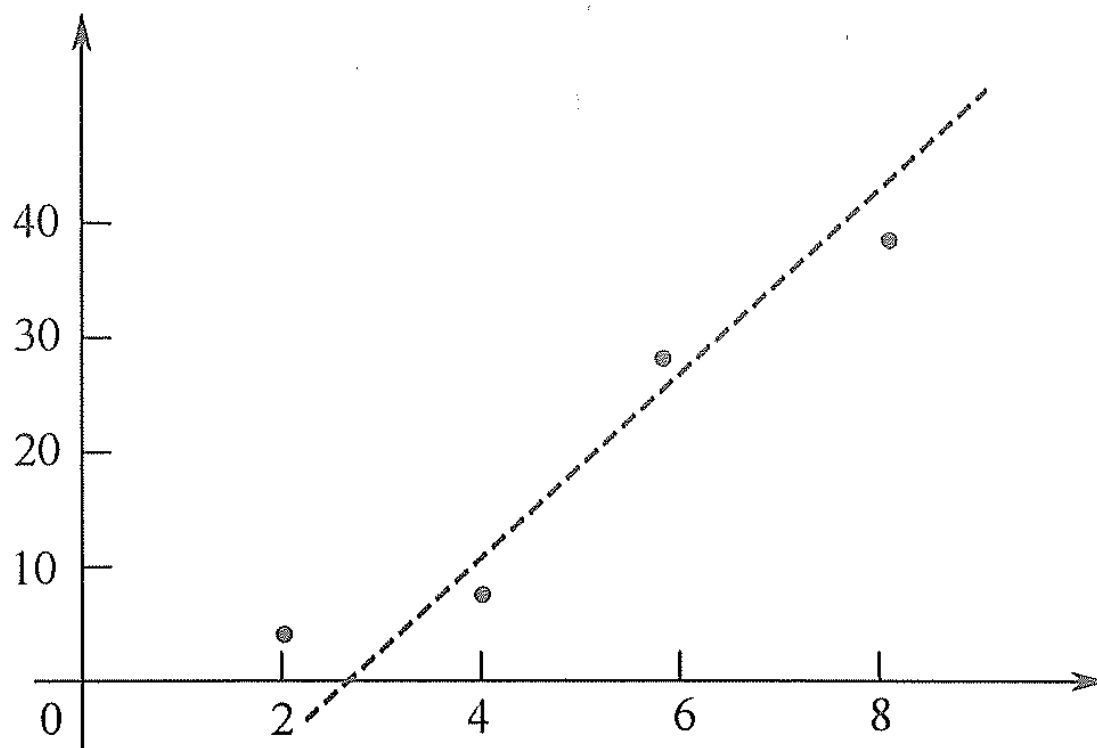
- 例7.5.1

- 例7.5.2

例 7.5.1 已知实验数据如下：求拟合曲线 $y = p(x)$ 。

i	0	1	2	3
x_i	2	4	6	8
y_i	2	11	28	40

(1) 在坐标平面上描出点 (x_i, y_i) ($i = 0, 1, 2, 3$),



(2) 根据各点的分布情况，选用线性函数

$$p(x) = a_0 + a_1 x \quad \text{故取 } \varphi_0(x) = 1, \varphi_1(x) = x。$$

(3) 建立法方程，

$$(\varphi_0, \varphi_0) = \sum 1 = 4, \quad (\varphi_0, \varphi_1) = (\varphi_1, \varphi_0) = \sum x_i = 20$$

$$(\varphi_1, \varphi_1) = \sum x_i^2 = 120, \quad (f, \varphi_0) = d_0 = \sum y_i = 81$$

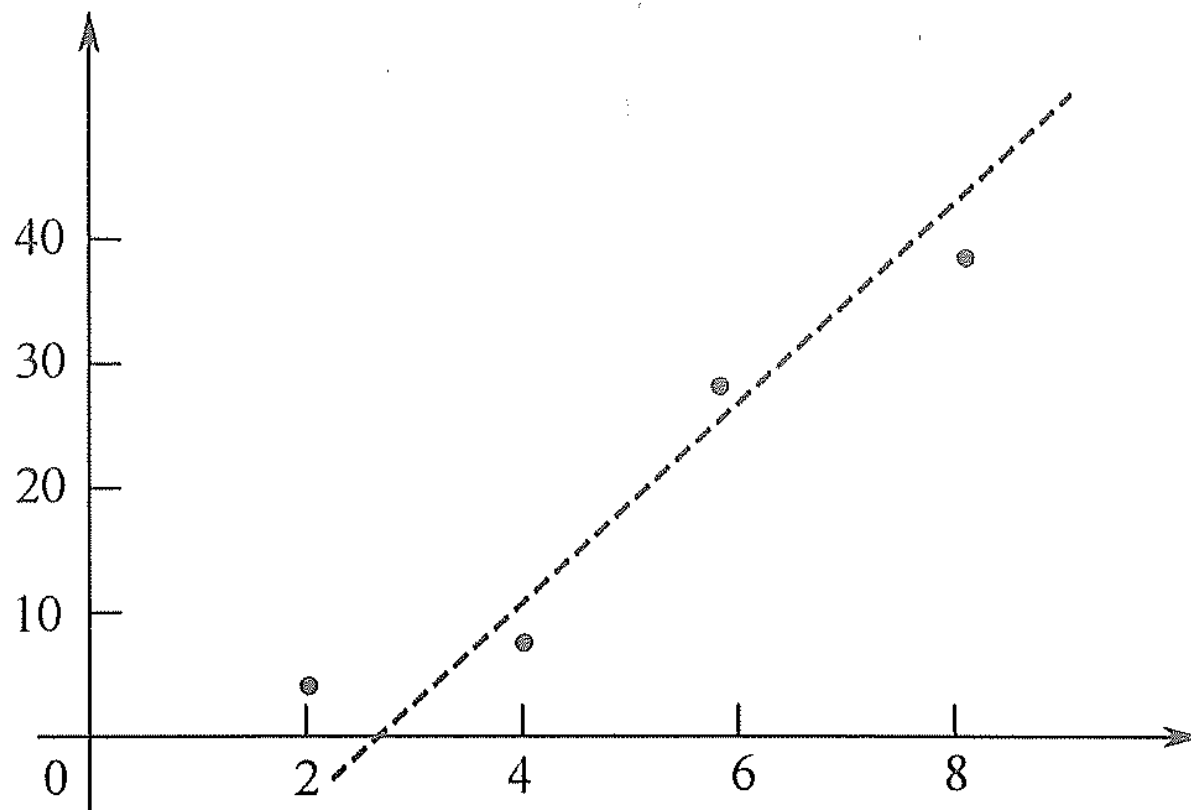
$$\begin{pmatrix} 4 & 20 \\ 20 & 120 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} 81 \\ 536 \end{pmatrix} \quad (f, \varphi_1) = d_1 = \sum x_i y_i = 536$$

$$a_0 = -12.5, \quad a_1 = 6.55$$

$$p(x) = -12.5 + 6.55x$$

$$\begin{bmatrix} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_1) & \cdots & (\varphi_0, \varphi_m) \\ (\varphi_1, \varphi_0) & (\varphi_1, \varphi_1) & \cdots & (\varphi_1, \varphi_m) \\ \cdots & \cdots & \cdots & \cdots \\ (\varphi_m, \varphi_0) & (\varphi_m, \varphi_1) & \cdots & (\varphi_m, \varphi_m) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \cdots \\ a_m \end{bmatrix} = \begin{bmatrix} d_0 \\ d_1 \\ \cdots \\ d_m \end{bmatrix}$$

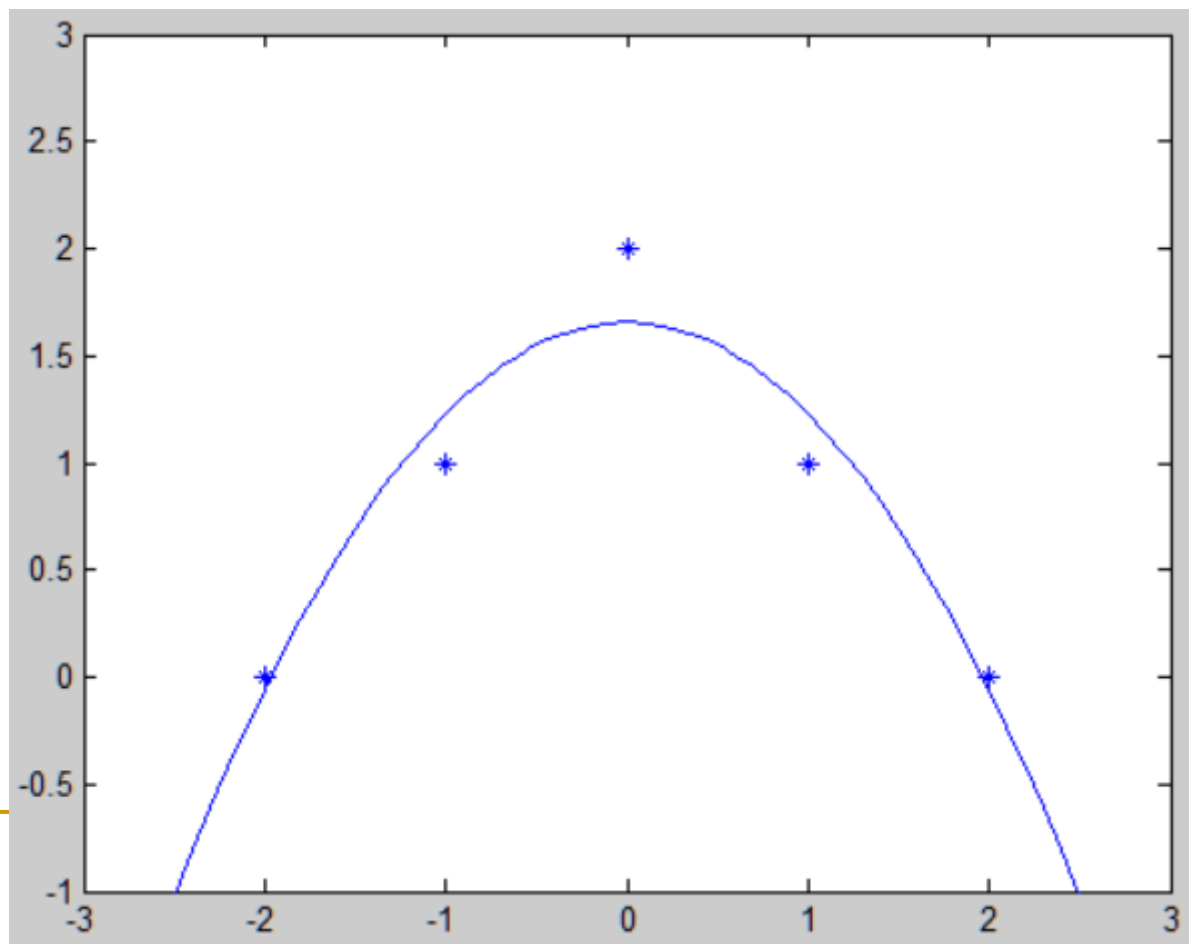
$$p(x) = -12.5 + 6.55x$$



例 7.5.2 已知函数表如下：

x	-2	-1	0	1	2
y	0	1	2	1	0

试用二次多项式 $p(x) = a_0 + a_1x + a_2x^2$ 拟合这组数据。



解

x	-2	-1	0	1	2
y	0	1	2	1	0

$$G = \begin{pmatrix} (1,1) & (1,x) & (1,x^2) \\ (x,1) & (x,x) & (x,x^2) \\ (x^2,1) & (x^2,x) & (x^2,x^2) \end{pmatrix} = \begin{pmatrix} 5 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{pmatrix}$$

法方程
系数矩阵

法方程的右端项为

$$((y,1), (y,x), (y,x^2))^T = (4, 0, 2)^T$$

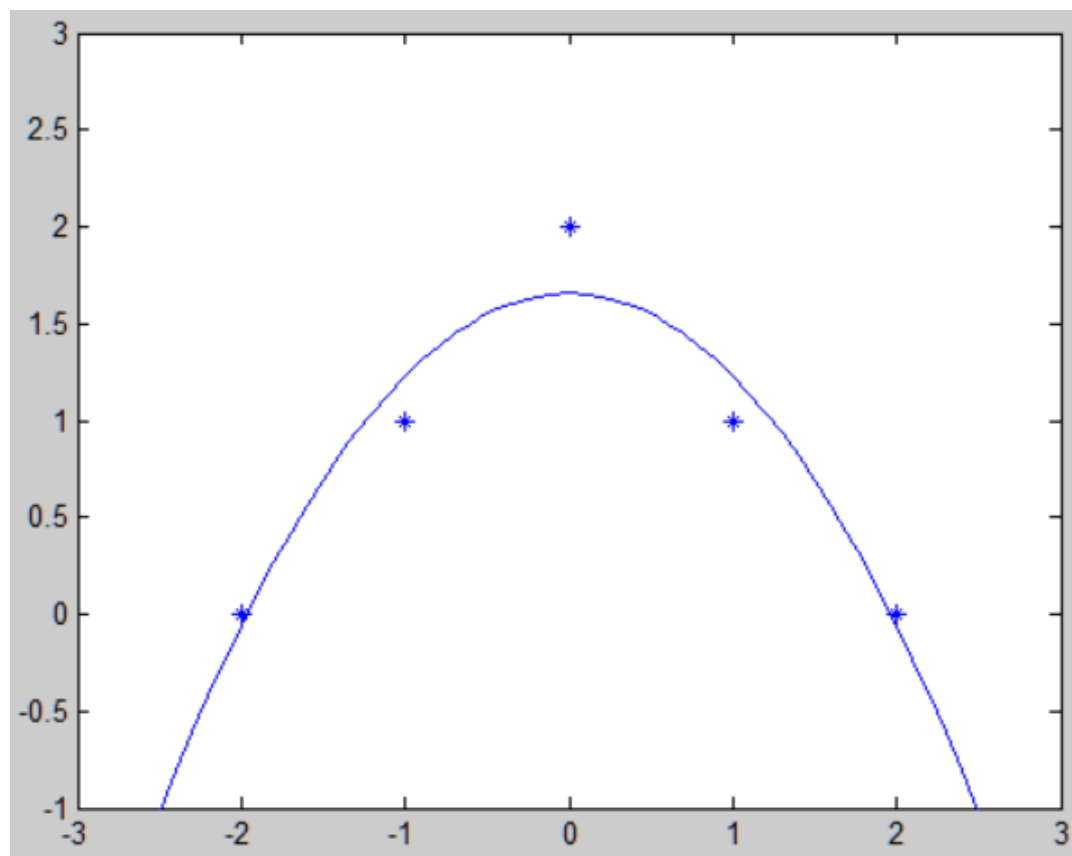
$$\begin{bmatrix} \sum 1 & \sum x_i & \dots & \sum x_i^m \\ \sum x_i & \sum x_i^2 & \dots & \sum x_i^{m-1} \\ \dots & \dots & \dots & \dots \\ \sum x_i^m & \sum x_i^{m+1} & \dots & \sum x_i^{2m} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_m \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \dots \\ \sum x_i^m y_i \end{bmatrix}$$

$$\begin{pmatrix} 5 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix}$$

$$a_0 = \frac{58}{35}, \quad a_1 = 0, \quad a_2 = -\frac{3}{7}$$

$$p(x) = \frac{58}{35} - \frac{3}{7}x^2$$

$$p(x) = \frac{58}{35} - \frac{3}{7}x^2$$



BACK

7.5.3 正交多项式拟合

$$\begin{bmatrix} \sum 1 & \sum x_i & \dots & \sum x_i^m \\ \sum x_i & \sum x_i^2 & \dots & \sum x_i^{m-1} \\ \dots & \dots & \dots & \dots \\ \sum x_i^m & \sum x_i^{m+1} & \dots & \sum x_i^{2m} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_m \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \dots \\ \sum x_i^m y_i \end{bmatrix} \quad (7.5.5)$$

由于当给定数据点 n 较大时，用多项式拟合时所求的法方程(7.5.5)式一般为“病态”方程组。

$$\begin{bmatrix} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_1) & \dots & (\varphi_0, \varphi_m) \\ (\varphi_1, \varphi_0) & (\varphi_1, \varphi_1) & \dots & (\varphi_1, \varphi_m) \\ \dots & \dots & \dots & \dots \\ (\varphi_m, \varphi_0) & (\varphi_m, \varphi_1) & \dots & (\varphi_m, \varphi_m) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_m \end{bmatrix} = \begin{bmatrix} d_0 \\ d_1 \\ \dots \\ d_m \end{bmatrix} \quad (7.5.4)$$

为了避免求解法方程(7.5.5)式，可以改用正交多项式，进行正交多项式拟合。

$$\begin{bmatrix} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_1) & \dots & (\varphi_0, \varphi_m) \\ (\varphi_1, \varphi_0) & (\varphi_1, \varphi_1) & \dots & (\varphi_1, \varphi_m) \\ \dots & \dots & \dots & \dots \\ (\varphi_m, \varphi_0) & (\varphi_m, \varphi_1) & \dots & (\varphi_m, \varphi_m) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_m \end{bmatrix} = \begin{bmatrix} d_0 \\ d_1 \\ \dots \\ d_m \end{bmatrix}$$

(7.5.4)

$$(\varphi_j, \varphi_k) = \sum_{i=0}^n \varphi_j(x_i) \varphi_k(x_i)$$

$$(f, \varphi_k) = \sum_{i=0}^n y_i \varphi_k(x_i) = d_k$$

多项式拟合



当函数类 $\Phi = \{\varphi_0, \varphi_1, \dots, \varphi_m\} = \{1, x, x^2, \dots, x^m\}$ 时：

$$p(x) = a_0 + a_1 x + \dots + a_m x^m$$

$$\begin{bmatrix} \sum 1 & \sum x_i & \dots & \sum x_i^m \\ \sum x_i & \sum x_i^2 & \dots & \sum x_i^{m-1} \\ \dots & \dots & \dots & \dots \\ \sum x_i^m & \sum x_i^{m+1} & \dots & \sum x_i^{2m} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_m \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \dots \\ \sum x_i^m y_i \end{bmatrix}$$

(7.5.5)

离散点集上的正交多项式

$$\{p_0(x), p_1(x), \dots, p_n(x)\}:$$

$$(p_k, p_j) = \sum_{i=0}^m \omega_i p_k(x_i) p_j(x_i)$$

$$\{x_i\} \quad (i = 0, 1, 2, \dots, m)$$

$$\{\omega_i\} \quad (i = 0, 1, 2, \dots, m)$$

若

$$(p_k, p_j) = \begin{cases} 0 & j \neq k \\ A_k & j = k \end{cases}$$

则称多项式族 $\{p_i(x)\} \quad (i = 0, 1, 2, \dots, n)$ 为关于离散节点 $\{x_i\} \quad (i = 0, 1, 2, \dots, m)$ 的带权 $\omega_i \quad (i = 0, 1, 2, \dots, m)$ 正交多项式族。

最高次项系数为 1(首 1) 的正交多项式族 $\{p_k(x)\}$ 有下面的递推公式:

$$\begin{cases} p_0(x) = 1 \\ p_1(x) = (x - \alpha_0)p_0(x) = x - \alpha_0 \\ p_{k+1}(x) = (x - \alpha_k)p_k(x) - \beta_{k-1}p_{k-1}(x) \end{cases}$$

其中:

$$\begin{cases} \alpha_k = \frac{(xp_k, p_k)}{(p_k, p_k)} \\ \beta_{k-1} = \frac{(p_k, p_k)}{(p_{k-1}, p_{k-1})} \end{cases} \quad (k = 1, 2, \dots, n)$$

例 7.5.3

构造关于离散点集 $\{-2, -1, 0, 1, 2\}$ ，权系数为 $\{1, 1, 1, 1, 1\}$ ，首项系数为 1 的正交多项式族 $p_0(x), p_1(x), p_2(x)$ 。

解

$$p_0(x) = 1$$

$$\alpha_0 = \frac{(xp_0, p_0)}{(p_0, p_0)} = \frac{0}{5} = 0$$

$$p_1(x) = (x - \alpha_0)p_0(x) = x$$

$$\alpha_1 = \frac{(xp_1, p_1)}{(p_1, p_1)} = \frac{0}{10} = 0$$

$$\beta_0 = \frac{(p_1, p_1)}{(p_0, p_0)} = \frac{10}{5} = 2$$

$$p_2(x) = (x - \alpha_1)p_1 - \beta_0 p_0 = x^2 - 2$$

$$\begin{cases} p_0(x) = 1 \\ p_1(x) = (x - \alpha_0)p_0(x) = x - \alpha_0 \\ p_{k+1}(x) = (x - \alpha_k)p_k(x) - \beta_{k-1}p_{k-1}(x) \end{cases}$$

$$\begin{cases} \alpha_k = \frac{(xp_k, p_k)}{(p_k, p_k)} \\ \beta_{k-1} = \frac{(p_k, p_k)}{(p_{k-1}, p_{k-1})} \quad (k = 1, 2, \dots, n) \end{cases}$$

$$(p_k, p_j) = \sum_{i=1}^m \omega_i p_k(x_i) p_j(x_i)$$

$$\begin{bmatrix} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_1) & \dots & (\varphi_0, \varphi_m) \\ (\varphi_1, \varphi_0) & (\varphi_1, \varphi_1) & \dots & (\varphi_1, \varphi_m) \\ \dots & \dots & \dots & \dots \\ (\varphi_m, \varphi_0) & (\varphi_m, \varphi_1) & \dots & (\varphi_m, \varphi_m) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_m \end{bmatrix} = \begin{bmatrix} d_0 \\ d_1 \\ \dots \\ d_m \end{bmatrix}$$

$$(\varphi_j, \varphi_k) = \sum_{i=0}^n \varphi_j(x_i) \varphi_k(x_i)$$

$$(f, \varphi_k) = \sum_{i=0}^n y_i \varphi_k(x_i) = d_k$$

(7.5.4)

多项式拟合



当函数类 $\Phi = \{\varphi_0, \varphi_1, \dots, \varphi_m\} = \{1, x, x^2, \dots, x^m\}$ 时：

$$p(x) = a_0 + a_1 x + \dots + a_m x^m$$

$$\begin{bmatrix} \sum 1 & \sum x_i & \dots & \sum x_i^m \\ \sum x_i & \sum x_i^2 & \dots & \sum x_i^{m-1} \\ \dots & \dots & \dots & \dots \\ \sum x_i^m & \sum x_i^{m+1} & \dots & \sum x_i^{2m} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_m \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \dots \\ \sum x_i^m y_i \end{bmatrix} \quad (7.5.5)$$

$$\begin{bmatrix} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_1) & \dots & (\varphi_0, \varphi_m) \\ (\varphi_1, \varphi_0) & (\varphi_1, \varphi_1) & \dots & (\varphi_1, \varphi_m) \\ \dots & \dots & \dots & \dots \\ (\varphi_m, \varphi_0) & (\varphi_m, \varphi_1) & \dots & (\varphi_m, \varphi_m) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_m \end{bmatrix} = \begin{bmatrix} d_0 \\ d_1 \\ \dots \\ d_m \end{bmatrix}$$

$$(\varphi_j, \varphi_k) = \sum_{i=0}^n \varphi_j(x_i) \varphi_k(x_i)$$

$$(f, \varphi_k) = \sum_{i=0}^n y_i \varphi_k(x_i) = d_k$$

正交多项式拟合

(7.5.4)

当函数类 $\Phi = \{\varphi_0, \varphi_1, \dots, \varphi_m\}$ 采用正交多项式族 $p_0(x), p_1(x), \dots, p_n(x)$ 时

$$(p_k, p_k) a_k = (f, p_k) \quad (k = 0, 1, 2, \dots, n)$$

$$a_k = \frac{(f, p_k)}{(p_k, p_k)} \quad (k = 0, 1, \dots, n)$$

$$p(x) = \sum_{k=0}^n a_k p_k(x)$$

例 7.5.4 用正交多项式族，求例 7.5.2 的拟合多项式。

x	-2	-1	0	1	2
y	0	1	2	1	0

试用二次多项式 $p(x) = a_0 + a_1x + a_2x^2$ 拟合这组数据。

解

(1) 建立关于节点 $\{-2, -1, 0, 1, 2\}$ 的正交多项式族，由例 7.5.3 可知

$$p_0(x) = 1, p_1(x) = x, p_2(x) = x^2 - 2$$

(2) 利用式 (7.5.8)，得二次拟合多项式为

$$p(x) = \frac{(y, p_0)}{(p_0, p_0)} p_0(x) + \frac{(y, p_1)}{(p_1, p_1)} p_1(x) + \frac{(y, p_2)}{(p_2, p_2)} p_2(x)$$

$$= \frac{4}{5} \times 1 + \frac{0}{10} \times x + \frac{-6}{14} (x^2 - 2) = \frac{58}{35} - \frac{3}{7} x^2$$

■ End!

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